

CO3-AT1 : Experiments

Python and R: Code and Output (Side by Side)

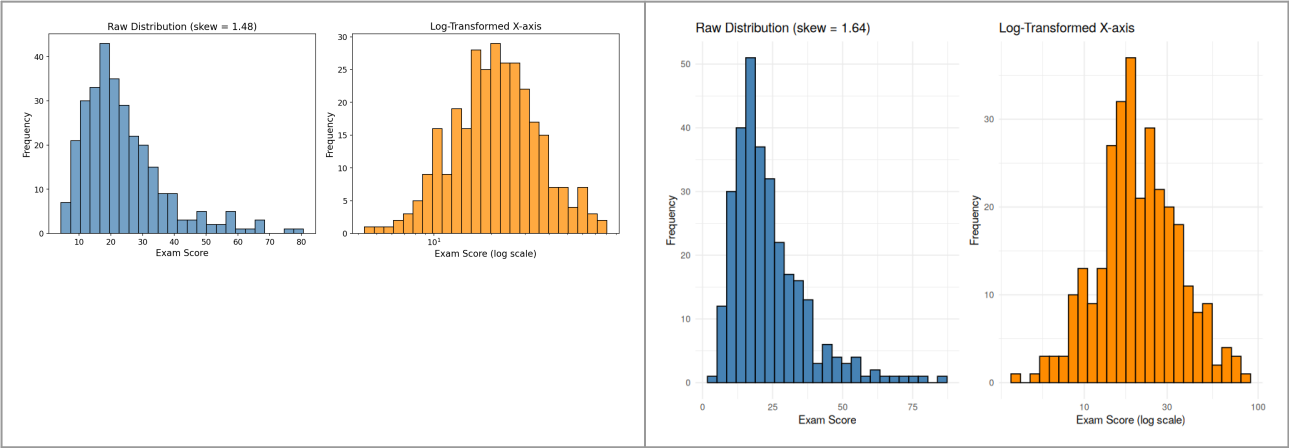
VISHAAL M S

Roll No: 192424420

Experiment 1

Highly Skewed Distributions

Python	R
<pre>import matplotlib.pyplot as plt import seaborn as sns import numpy as np from scipy.stats import skew # (a) Simulate board-exam scores across 10 schools, # strongly right-skewed # (log-normal generator: most students score # low-to-moderate, a smaller # number score very high, producing a long right # tail). np.random.seed(1) schools = [f"School {i+1}" for i in range(10)] scores = np.random.lognormal(mean=3.0, sigma=0.55, size=300) scores = np.clip(scores, 1, 100) # (b) Raw distribution + skewness coefficient skew_val = skew(scores) print(f"Skewness (raw scores): {skew_val:.3f}") fig, axes = plt.subplots(1, 2, figsize=(12, 5)) sns.histplot(scores, bins=25, color='steelblue', ax=axes[0]) axes[0].set_title(f"Raw Distribution (skew = {skew_val:.2f})") axes[0].set_xlabel("Exam Score"); axes[0].set_ylabel("Frequency") sns.histplot(scores, bins=25, color='darkorange', log_scale=True, ax=axes[1]) axes[1].set_title("Log-Transformed X-axis") axes[1].set_xlabel("Exam Score (log scale)"); axes[1].set_ylabel("Frequency") plt.tight_layout() plt.show() """ (c) Which representation should the District Education Officer use, and why: The raw chart (skew ~ +1.48, strongly right-skewed) is dominated visually by a tall bar of low-to-mid scores with a long, sparse tail of high scorers stretching far to the right. In a board presentation, that long flat tail is easy to dismiss as "a few outliers" - understating how large and meaningful that top-scoring group actually is, since the raw x-axis compresses most of the visual detail into the leftmost bars. The LOG-TRANSFORMED chart is the one the DEO should use: on a log x-axis the lognormal data becomes close to symmetric/bell-shaped, so genuine structure (how spread out typical performance is, where the bulk of students actually sit) becomes readable instead of hidden in a cramped left-hand spike. The untransformed chart could mislead a non-technical</pre>	<pre>library(ggplot2) library(e1071) library(gridExtra) # (a) Simulate board-exam scores across 10 schools, # strongly right-skewed set.seed(1) scores <- rlnorm(300, meanlog = 3.0, sdlog = 0.55) scores <- pmin(pmax(scores, 1), 100) df <- data.frame(Score = scores) # (b) Raw distribution + skewness coefficient skew_val <- skewness(scores) cat(sprintf("Skewness (raw scores): %.3f\n", skew_val)) p1 <- ggplot(df, aes(x = Score)) + geom_histogram(bins = 25, fill = "steelblue", color = "black") + labs(title = sprintf("Raw Distribution (skew = %.2f)", skew_val), x = "Exam Score", y = "Frequency") + theme_minimal() p2 <- ggplot(df, aes(x = Score)) + geom_histogram(bins = 25, fill = "darkorange", color = "black") + scale_x_log10() + labs(title = "Log-Transformed X-axis", x = "Exam Score (log scale)", y = "Frequency") + theme_minimal() grid.arrange(p1, p2, ncol = 2) # (c) Which representation should the District # Education Officer use, and why: # The raw chart (skew ~ +1.64, strongly right-skewed) # is dominated visually # by a tall bar of low-to-mid scores with a long, # sparse tail of high # scorers stretching far to the right. In a board # presentation, that long # flat tail is easy to dismiss as "a few outliers" - # understating how large # and meaningful that top-scoring group actually is, # since the raw x-axis # compresses most of the visual detail into the # leftmost bars. # The LOG-TRANSFORMED chart is the one the DEO should # use: on a log x-axis # the lognormal data becomes close to # symmetric/bell-shaped, so genuine # structure (how spread out typical performance is, # where the bulk of # students actually sit) becomes readable instead of # hidden in a cramped # left-hand spike. The untransformed chart could # mislead a non-technical # audience into underestimating both how skewed # performance is and how # many students sit far from the "typical" score, # since visual bar height</pre>
<pre>audience into underestimating both how skewed performance is and how many students sit far from the "typical" score, since visual bar height differences on a linear axis don't correspond to equal proportional differences the way they do on the log axis. """</pre>	<pre># differences on a linear axis don't correspond to equal proportional # differences the way they do on the log axis.</pre>
Output	Output



Experiment 2

Bean Plots

Python	R
<pre>import matplotlib.pyplot as plt import seaborn as sns import numpy as np import pandas as pd # (a) Simulate monthly data usage (GB) per subscriber, # 4 plan-tier groups # within 8 service circles, and build a bean-plot # approximation using # seaborn's violinplot() + stripplot() combined on the # same axes. np.random.seed(2) circles = [f"Circle {i+1}" for i in range(8)] tiers = ['Basic', 'Standard', 'Premium', 'Unlimited'] tier_mean = {'Basic': 4, 'Standard': 10, 'Premium': 20, 'Unlimited': 35} rows = [] for circle in circles: for tier in tiers: usage = np.random.normal(tier_mean[tier], tier_mean[tier] * 0.3, 15) for u in np.clip(usage, 0.5, None): rows.append([circle, tier, u]) df = pd.DataFrame(rows, columns=['ServiceCircle', 'Tier', 'DataUsageGB']) plt.figure(figsize=(12, 6)) sns.violinplot(data=df, x='ServiceCircle', y='DataUsageGB', inner=None, color='lightblue', linewidth=1, cut=0) sns.stripplot(data=df, x='ServiceCircle', y='DataUsageGB', hue='Tier', dodge=False, alpha=0.6, size=3, jitter=0.25) plt.title("Monthly Data Usage per Subscriber by Service Circle (Bean-Plot Approximation)") plt.xlabel("Service Circle"); plt.ylabel("Data Usage (GB)") plt.legend(title="Plan Tier", bbox_to_anchor=(1.02, 1), loc='upper left') plt.tight_layout() plt.show() # (b) Compare to a standard boxplot plt.figure(figsize=(10, 5.5)) sns.boxplot(data=df, x='ServiceCircle', y='DataUsageGB', color='lightgreen') plt.title("Monthly Data Usage per Subscriber by Service Circle (Standard Boxplot)") plt.xlabel("Service Circle"); plt.ylabel("Data Usage (GB)") plt.tight_layout() plt.show() """ (b continued) What the bean plot reveals that the boxplot hides: Because each service circle actually contains 4 distinct usage tiers (Basic/Standard/Premium/Unlimited) simulated around very different</pre>	<pre>library(ggplot2) library(ggbeeswarm) # (a) Simulate monthly data usage (GB) per subscriber, # 4 plan-tier groups # within 8 service circles. R's dedicated `beanplot` # package is not # installable in this environment (no CRAN access), so # a violin + # quasirandom-point combination (ggplot2 + ggbeeswarm) # is used instead - # the same density-plus-individual-points # approximation the assignment # itself allows as a Python fallback. set.seed(2) circles <- paste0("Circle ", 1:8) tiers <- c('Basic', 'Standard', 'Premium', 'Unlimited') tier_mean <- c(Basic=4, Standard=10, Premium=20, Unlimited=35) df <- data.frame() for (circle in circles) { for (tier in tiers) { usage <- rnorm(15, tier_mean[[tier]], tier_mean[[tier]] * 0.3, 0.5) df <- rbind(df, data.frame(ServiceCircle=circle, Tier=tier, DataUsageGB=usage)) } } p1 <- ggplot(df, aes(x=ServiceCircle, y=DataUsageGB)) + geom_violin(fill="lightblue", trim=TRUE) + geom_quasirandom(aes(color=Tier), size=1, alpha=0.7, width=0.25) + labs(title="Monthly Data Usage per Subscriber by Service Circle (Bean-Plot Approximation)", x="Service Circle", y="Data Usage (GB)") + theme_minimal() + theme(axis.text.x=element_text(angle=30, hjust=1)) print(p1) # (b) Compare to a standard boxplot p2 <- ggplot(df, aes(x=ServiceCircle, y=DataUsageGB)) + geom_boxplot(fill="lightgreen") + labs(title="Monthly Data Usage per Subscriber by Service Circle (Standard Boxplot)", x="Service Circle", y="Data Usage (GB)") + theme_minimal() + theme(axis.text.x=element_text(angle=30, hjust=1)) print(p2) # (b continued) What the bean plot reveals that the # boxplot hides: # Because each service circle actually contains 4 # distinct usage tiers # (Basic/Standard/Premium/Unlimited) simulated around # very different # means (4, 10, 20, 35 GB), the true per-circle # distribution is</pre>

means (4, 10, 20, 35 GB), the true per-circle distribution is MULTIMODAL - four separate clusters of points. The boxplot collapses this entirely into a single box (median, IQR, whiskers), which looks like one smooth, unimodal spread and completely hides the four-tier structure. The bean/strip view shows the actual clumps of individual points directly, making the multimodality immediately visible.

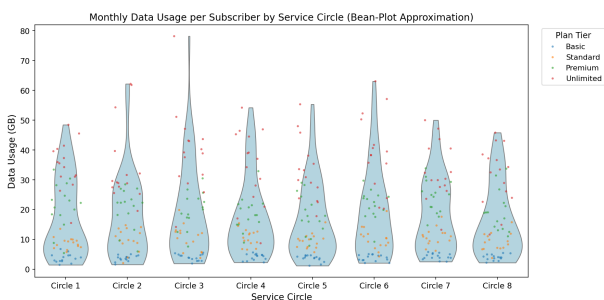
(c) Recommendation to the Network Operations Head: For an internal report aimed at a NON-TECHNICAL audience, the standard BOXPLOT is the better choice. It is far faster to read (median, typical range, and outliers in one glance) and doesn't require the audience to interpret a dense cloud of overlapping points and violin shapes across 8 circles at once, which can look cluttered and intimidating to readers unfamiliar with density-based visualizations. The bean plot is more appropriate for a technical/analyst audience that specifically needs to see the tier-driven multimodal structure the boxplot would otherwise hide - it should be reserved for a technical appendix rather than the main non-technical report.

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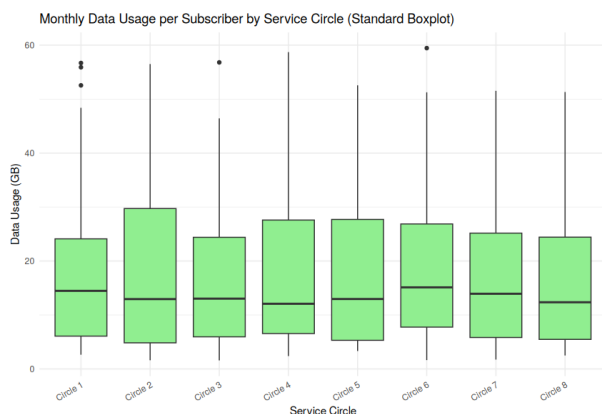
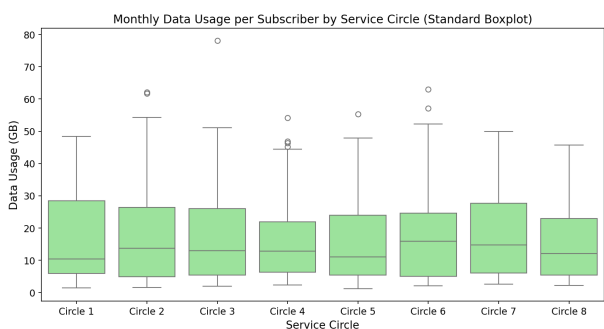
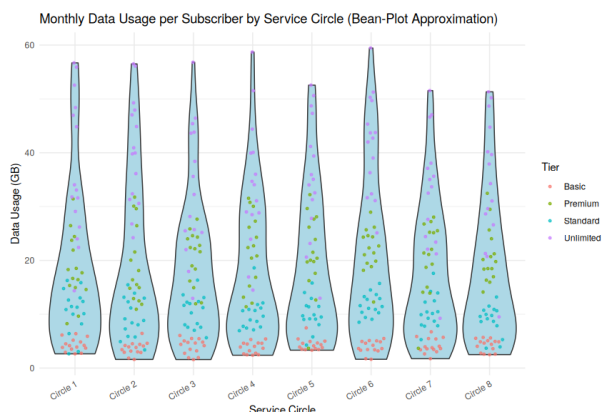
MULTIMODAL - four separate clusters of points. The boxplot collapses this entirely into a single box (median, IQR, whiskers), which looks like one smooth, unimodal spread and completely hides the four-tier structure. The bean/strip view shows the actual clumps of individual points directly, making the multimodality immediately visible.

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Output



Output



Experiment 3

Treemaps

Python	R
<pre>import matplotlib.pyplot as plt import squarify import pandas as pd import numpy as np # (a) Compile board-exam scores for 10 schools, with a # second-level # breakdown (Science / Arts / Commerce streams) within # each school, sized # by total aggregate score. np.random.seed(3) schools = [f"School {i+1}" for i in range(10)] streams = ['Science', 'Arts', 'Commerce'] rows = [] for school in schools: for stream in streams: total_score = np.random.randint(500, 3000) rows.append([school, stream, total_score]) df = pd.DataFrame(rows, columns=['School', 'Stream', 'TotalScore']) school_totals = df.groupby('School')['TotalScore'].sum() ().sort_values(ascending=False) plt.figure(figsize=(10, 6)) labels = [f"{s}\n{v:}" for s, v in school_totals.items()] squarify.plot(sizes=school_totals.values, label=labels, color=plt.cm.tab10.colors, edgecolor='white', linewidth=1.5) plt.title("Aggregate Board-Exam Scores by School (Treemap)") plt.axis('off') plt.tight_layout() plt.show() # (b) Which school dominates the treemap by area? top_school = school_totals.index[0] top_val = school_totals.iloc[0] total_all = school_totals.sum() print(f"Dominant school: {top_school} with {top_val:}, {top_val/total_all:.1%} of total") """ (b continued) Is this dominance proportionally accurate or visually exaggerated? Treemaps size each rectangle by AREA, which is proportional to value - this is proportionally accurate in principle, but human perception of area is known to under- or over-estimate differences compared to perception of length (as in a bar chart). A rectangle that is, say, 40% larger in value can visually appear more than 40% larger to the eye because we don't judge 2D area changes as precisely as 1D length changes - so while the encoding itself is</pre>	<pre>library(ggplot2) library(dplyr) # (a) Compile board-exam scores for 10 schools, sized # by total aggregate # score. Built manually with geom_rect (a squarified # single-level layout) # since the treemapify / treemap CRAN packages are not # installable in # this environment (no CRAN access). set.seed(3) schools <- paste0("School ", 1:10) streams <- c('Science', 'Arts', 'Commerce') df <- data.frame() for (school in schools) { for (stream in streams) { total_score <- sample(500:3000, 1) df <- rbind(df, data.frame(School=school, Stream=stream, TotalScore=total_score)) } } school_totals <- df %>% group_by(School) %>% summarise(Total=sum(TotalScore)) %>% arrange(desc(Total)) %>% mutate(frac = Total / sum(Total), xmax = cumsum(frac), xmin = lag(xmax, default=0)) p <- ggplot(school_totals) + geom_rect(aes(xmin=xmin, xmax=xmax, ymin=0, ymax=1, fill=School), color="white") + geom_text(aes(x=(xmin+xmax)/2, y=0.5, label=paste0(School, "\n", format(Total, big.mark=","))), size=3) + labs(title="Aggregate Board-Exam Scores by School (Treemap)") + theme_void() + theme(legend.position="none") print(p) # (b) Which school dominates the treemap by area? top_school <- school_totals\$School[1] top_val <- school_totals\$Total[1] total_all <- sum(school_totals\$Total) cat(sprintf("Dominant school: %s with %s (%.1f%% of total)\n", top_school, format(top_val, big.mark=","), 100 * top_val / total_all)) # (b continued) Is this dominance proportionally # accurate or visually # exaggerated? # Treemaps size each rectangle by AREA, which is # proportional to value - # this is proportionally accurate in principle, but # human perception of # area is known to under- or over-estimate differences # compared to # perception of length (as in a bar chart). A # rectangle that is, say, # 40% larger in value can visually appear more than # 40% larger to the</pre>

mathematically fair, the PERCEIVED dominance of the top school can feel slightly exaggerated compared to what a bar chart of the same numbers would convey.

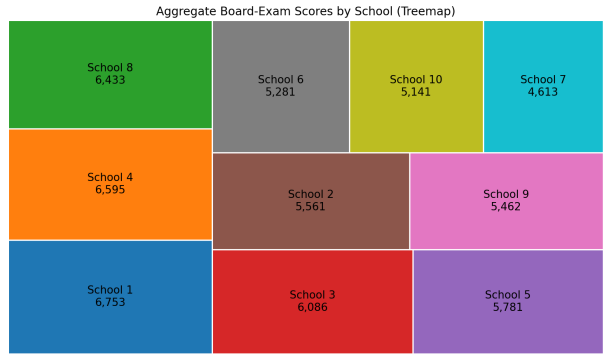
(c) Treemap vs bar chart - one advantage when a second-level breakdown is included:
A simple bar chart of school totals can only show ONE level (total score per school) - adding the stream-level breakdown would require either a stacked or grouped bar chart, which becomes visually busy with 10 schools x 3 streams = 30 bars/segments. A treemap can nest the second level directly inside each school's rectangle (sub-rectangles for Science/Arts/Commerce), letting a viewer see both the school-level totals AND the internal composition in the same compact chart, without needing a separate legend-heavy multi-bar layout.

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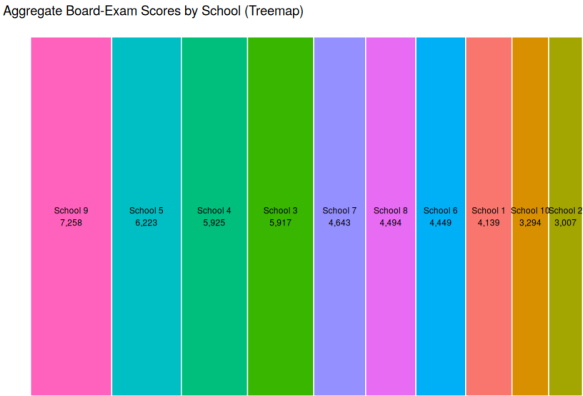
eye because we don't judge 2D area changes as precisely as 1D length
changes - so while the encoding itself is mathematically fair, the
PERCEIVED dominance of the top school can feel slightly exaggerated
compared to what a bar chart of the same numbers would convey.

(c) Treemap vs bar chart - one advantage when a second-level breakdown is included:
A simple bar chart of school totals can only show ONE level (total score per school) - adding the stream-level breakdown would require either a stacked or grouped bar chart, which becomes visually busy
with 10 schools x 3 streams = 30 bars/segments. A treemap can nest the second level directly inside each school's rectangle (sub-rectangles for Science/Arts/Commerce), letting a viewer see both the school-level totals AND the internal composition in the same compact chart, without needing a separate legend-heavy multi-bar layout.

Output



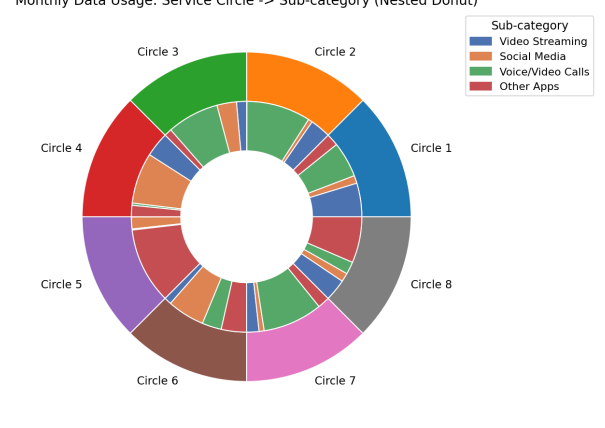
Output



Experiment 4

Visualizing Nested Proportions

Python	R
<pre>import matplotlib.pyplot as plt import numpy as np import pandas as pd # (a) Compile a nested (two-level) breakdown of monthly data usage (GB): # outer level = service circle (8), inner level = usage sub-category # (Voice-app / Video-streaming / Social-media / Other) within each circle. # Built as a matplotlib nested donut chart (two concentric pie rings), # which directly shows nested part-to-whole proportions. np.random.seed(4) circles = [f"Circle {i+1}" for i in range(8)] subcats = ['Video Streaming', 'Social Media', 'Voice/Video Calls', 'Other Apps'] data = {} for circle in circles: weights = np.random.dirichlet(np.ones(4)) # ensures each circle's shares sum to 1 data[circle] = dict(zip(subcats, weights)) outer_values = [1] * len(circles) # equal-width outer ring segments (one per circle) inner_values = [] inner_colors = [] subcat_colors = {'Video Streaming': '#4C72B0', 'Social Media': '#DD8452', 'Voice/Video Calls': '#55A868', 'Other Apps': '#C44E52'} for circle in circles: for subcat in subcats: inner_values.append(data[circle][subcat]) inner_colors.append(subcat_colors[subcat]) outer_colors = plt.cm.tab10.colors[:len(circles)] fig, ax = plt.subplots(figsize=(8, 8)) ax.pie(outer_values, radius=1.0, colors=outer_colors, labels=circles, wedgeprops=dict(width=0.3, edgecolor='white'), labeldistance=1.08) ax.pie(inner_values, radius=0.7, colors=inner_colors, wedgeprops=dict(width=0.3, edgecolor='white')) ax.set_title("Monthly Data Usage: Service Circle -&gt; Sub-category (Nested Donut)") handles = [plt.Rectangle((0, 0), 1, 1, color=c) for c in subcat_colors.values()] ax.legend(handles, subcat_colors.keys(), title="Sub-category", loc='upper left', bbox_to_anchor=(1.02, 1)) plt.tight_layout() plt.show() # (b) Verify programmatically that inner-level percentages sum to 100% per circle print("Inner-level percentage check (should all be 100.0%):") for circle in circles:</pre>	<pre>library(ggplot2) library(dplyr) # (a) Compile a nested (two-level) breakdown of monthly data usage (GB): # outer level = service circle (8), inner level = usage sub-category # within each circle. Built with ggplot2 faceted donut charts (coord_polar charts (coord_polar # per circle), which clearly shows each circle's internal proportions # side by side. set.seed(4) circles &lt;- paste0("Circle ", 1:8) subcats &lt;- c('Video Streaming', 'Social Media', 'Voice/Video Calls', 'Other Apps') df &lt;- data.frame() for (circle in circles) { raw &lt;- runif(4) shares &lt;- raw / sum(raw) # ensures each circle's shares sum to 1 df &lt;- rbind(df, data.frame(ServiceCircle=circle, SubCategory=subcats, Share=shares)) } p &lt;- ggplot(df, aes(x=2, y=Share, fill=SubCategory)) + geom_col(color="white") + coord_polar(theta="y") + xlim(0.5, 2.5) + facet_wrap(~ServiceCircle, ncol=4) + labs(title="Monthly Data Usage: Service Circle -&gt; Sub-category (Faceted Donut)", fill="Sub-category") + theme_void() + theme(strip.text=element_text(size=10, face="bold")) print(p) # (b) Verify programmatically that inner-level percentages sum to 100% per circle check &lt;- df %>% group_by(ServiceCircle) %>% summarise(TotalPct = sum(Share) * 100) cat("Inner-level percentage check (should all be 100.0%):\n") print(check) # (c) Interpretation risk of nested-proportion charts: # A key risk is comparing ANGLES or AREAS across DIFFERENT outer groups - # e.g. a viewer might try to compare "Video Streaming's share in Circle 1" # directly against "Video Streaming's share in Circle 5" just by eyeballing # wedge angle/area, but each donut here only encodes WITHIN-circle # proportions, not across-circle absolute usage volume. This faceted # design mitigates the risk by drawing every circle's donut at the same # fixed size regardless of that circle's actual total usage, and by using # consistent sub-category colors across all panels -</pre>

<pre>total_pct = sum(data[circle].values()) * 100 print(f" {circle}: {total_pct:.1f}%") """ (c) Interpretation risk of nested-proportion charts: A key risk is comparing ANGLES or AREAS across DIFFERENT outer groups - e.g. a viewer might try to compare "Video Streaming's share in Circle 1" directly against "Video Streaming's share in Circle 5" just by eyeballing wedge angle/area, but because each circle's outer segment here is drawn with equal width (not sized by that circle's total usage), the inner wedge angles only encode WITHIN-circle proportions, not across-circle absolute usage. This chart design mitigates the risk partly by keeping outer segments equal-sized (so there's no illusion of comparing overall circle volume) and by using consistent sub-category colors across all circles, which at least makes proportion-vs-proportion comparison (not absolute-value comparison) the correct and intended reading of the chart. If absolute cross-circle comparison were needed, a faceted bar chart would be the safer alternative. """</pre>	<pre>so the only valid # comparison the chart invites is proportion-vs-proportion within a # circle, not absolute usage across circles. If absolute cross-circle # comparison were needed, a faceted bar chart would be the safer # alternative.</pre>
Output Monthly Data Usage: Service Circle -> Sub-category (Nested Donut) 	Output Monthly Data Usage: Service Circle -> Sub-category (Faceted Donut) 

Experiment 5

Quantile-Quantile (Q-Q) Plots

Python	R
<pre>import matplotlib.pyplot as plt import numpy as np from scipy import stats # (a) Simulate monthly data usage (GB) per subscriber # for 8 service # circles - drawn from a right-skewed (gamma) # distribution, since real # data usage is rarely normally distributed (heavy # users create a long # right tail), then build a Q-Q plot against a Normal # reference. np.random.seed(5) usage = np.random.gamma(shape=2.5, scale=6, size=200) fig, ax = plt.subplots(figsize=(7, 6)) stats.probplot(usage, dist="norm", plot=ax) ax.set_title("Q-Q Plot: Monthly Data Usage vs Normal Distribution") ax.get_lines()[0].set_markerfacecolor('steelblue') ax.get_lines()[0].set_markeredgecolor('black') ax.get_lines()[1].set_color('red') plt.tight_layout() plt.show() """ (b) Does the data deviate from normality, and where? The points follow the red reference line reasonably well through the LOWER-MIDDLE quantiles, but curve noticeably ABOVE the line at the upper-right end of the plot (the highest-usage subscribers). This upward curve in the upper tail is the classic Q-Q signature of a RIGHT-SKEWED distribution: the highest theoretical quantiles predicted under a Normal assumption are much lower than the actual highest observed usage values, meaning the real data has a heavier/longer right tail (a handful of very high-usage subscribers) than a Normal distribution would produce. The lower tail and center, by contrast, sit close to the line, so the deviation from normality is concentrated specifically in the upper tail rather than across the whole distribution. (c) Recommendation to the Network Operations Head: A statistical test or model that ASSUMES NORMALITY (e.g. a plain t-test or an unadjusted linear regression on raw usage) would NOT be appropriate here. The clear upper-tail departure seen in the Q-Q plot means normality-based confidence intervals or p-values would likely be distorted by the small number of extreme high-usage subscribers. A more</pre>	<pre># (a) Simulate monthly data usage (GB) per subscriber # for 8 service # circles - drawn from a right-skewed (gamma) # distribution, then build a # Q-Q plot against a Normal reference using base R's # qqnorm()/qqline(). set.seed(5) usage <- rgamma(200, shape = 2.5, scale = 6) qqnorm(usage, main = "Q-Q Plot: Monthly Data Usage vs Normal Distribution", pch = 19, col = "steelblue") qqline(usage, col = "red", lwd = 2) # (b) Does the data deviate from normality, and where? # The points follow the red reference line reasonably # well through the # lower-middle quantiles, but curve noticeably ABOVE # the line at the # upper-right end of the plot (the highest-usage # subscribers). This # upward curve in the upper tail is the classic Q-Q # signature of a # RIGHT-SKEWED distribution: the highest theoretical # quantiles predicted # under a Normal assumption are much lower than the # actual highest # observed usage values, meaning the real data has a # heavier/longer # right tail (a handful of very high-usage # subscribers) than a Normal # distribution would produce. The lower tail and # center, by contrast, # sit close to the line, so the deviation from # normality is concentrated # specifically in the upper tail rather than across # the whole # distribution. # (c) Recommendation to the Network Operations Head: # A statistical test or model that ASSUMES NORMALITY # (e.g. a plain t-test # or an unadjusted linear regression on raw usage) # would NOT be # appropriate here. The clear upper-tail departure # seen in the Q-Q plot # means normality-based confidence intervals or # p-values would likely be # distorted by the small number of extreme high-usage # subscribers. A more # appropriate approach would be to either transform # the data (e.g. a log # transform, which tends to pull in right-skewed # tails - similar to # Experiment 1) before applying normal-theory methods, # or to use a # distribution-appropriate model directly (e.g. a # Gamma or log-normal # generalized linear model), or a non-parametric # method that doesn't # assume normality at all.</pre>
<pre>appropriate approach would be to either transform the data (e.g. a log transform, which tends to pull in right-skewed tails - similar to Experiment 1) before applying normal-theory methods, or to use a distribution-appropriate model directly (e.g. a Gamma or log-normal generalized linear model), or a non-parametric method that doesn't assume normality at all. """</pre>	
Output	Output

